

Ervin Vavla

Curriculum Vitae



ABOUT ME

I have been passionate about electronics since I was young, and I chose my studies to better understand how electronic systems and circuits work. I enjoy learning about RF and analog circuits and testing my ideas with simulation tools such as LTSpice and ADS. I work well in a team and I am committed to contributing to technical goals and meeting deadlines.

PROFESSIONAL EXPERIENCE

MAY 2025 – OCTOBER 2025 (INTERNSHIP)

IHP Microelectronics - Germany *RF/mm-Wave Circuit Designer*

Designed and simulated an active balun in D-band using IHP SG13G3 SiGe BiCMOS technology, achieving low amplitude and phase imbalance, with 53 mW power consumption from a 1.8 V supply.

EXTRACURRICULAR EXPERIENCE

FEB 2024 – FEB 2025

Red Propulsion *Circuit Designer*

Designed an RF distribution system for two antenna types operating at 860 MHz and 2.4 GHz. Due to strict space constraints inside the rocket, I adapted the layout of three-way and six-way Wilkinson power splitter/combiners.

NOV 2023 – FEB 2024

Red Propulsion *GUI developer*

Learned Qt Creator and C++ to develop the graphical user interface.

EDUCATION

09/2022– 04/2026	Master in Electronic Systems Engineering Radio frequency applications <i>University of Florence</i> THESIS: DESIGN AND ANALYSIS OF A D-BAND (110-170 GHz) ACTIVE BALUN WITH LOW AMPLITUDE AND PHASE IMBALANCE IN 130 NM SIGE BICMOS TECHNOLOGY 110/110
2017 – 2021	Bachelor in Electronics and Telecommunications Engineering Electronic systems <i>University of Florence</i> THESIS: VHDL FOR REAL-TIME RECONFIGURATION OF THE PLL OPERATING FREQUENCY IN AN FPGA
2013 – 2017	High School Diploma Electronics <i>I.I.S Leonardo Da Vinci</i>

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PROJECTS

- 2021 **Basic signal generator**
Personal project
Implemented with Arduino AD9833 module
Prototyped and tested on breadboard
- 2017 **Cable tester**
I.I.S Leonardo Da Vinci - Final project
Designed with Multisim and Multiboard
Developed a cable tester capable of detecting open and short circuits in multi-conductor cables. In case of short circuits, the system identified the specific faulty conductors.
- 2016 **Analog stepper motor driver**
I.I.S Leonardo Da Vinci
Designed and realized with Multisim and Multiboard

COMPUTER SKILLS

INTERMEDIATE: $\LaTeX$, ADS, Cadence Virtuoso, LtSpice
BEGINNER: CST, VHDL, Fusion360, Altium, Multisim, Multiboard

LANGUAGES

ITALIAN	A1	●●●●●●●●	C2
ENGLISH	A1	●●●●●○	C2
ALBANIAN	A1	●●●○	C2

May 22, 2026

I authorize the processing of my personal data in accordance with Legislative Decree 196 of June 30, 2003, and Article 13 of the GDPR.

Vavla Ervin